

PRESENTATION OF FINDINGS

**Customer Profiling Through Transaction Behavior
for Marketing Strategy Optimization:
An Unsupervised Learning Approach
Using EDA, K-Means, and PCA**

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Presenter Introduction

CUSTOMER PROFILING THROUGH TRANSACTION BEHAVIOR

Using Data to Optimize Marketing Strategy

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Hello my name is Hannah Rose Bergman. I'm a grad student here at WGU pursuing my MSDA. I have a background in marketing analytics, specifically within the field of advertising and digital marketing, which is why I wanted the focus of this project to be on optimizing marketing strategy via customer profiling. Today we'll be looking at how we used customer transaction data to better understand our credit card users and create more effective marketing strategies. We'll walk through the problem we set out to solve, how we approached it, what we found and how these insights can help us take targeted action moving forward.

Problem Statement & Hypothesis

PROBLEM STATEMENT

- Current marketing is broad and untargeted, missing opportunities to personalize.
- Identify distinct customer profiles to improve targeting and engagement.

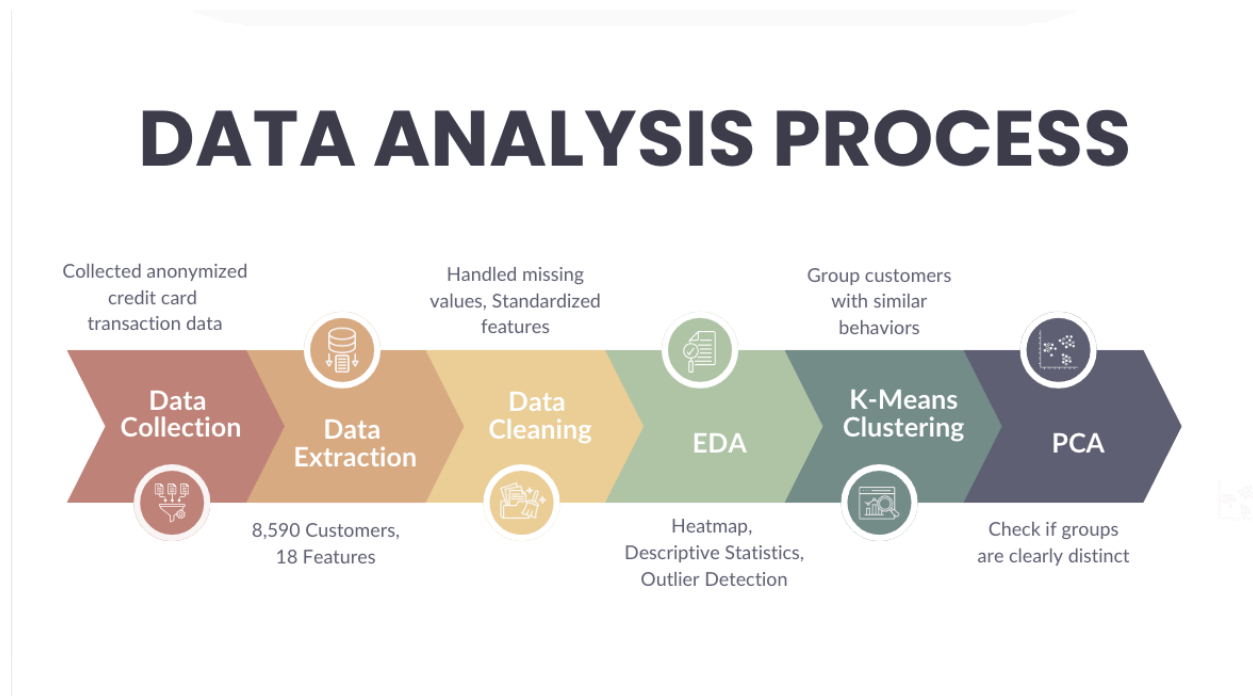


HYPOTHESIS

- H_0 : Segments will not be meaningfully distinct (silhouette score < 0.5)
- H_1 : Segments will be meaningfully distinct (silhouette score ≥ 0.5)

Our starting point was a common challenge - our current marketing was found to be too broad and untargeted. This meant that we were missing opportunities to connect with customers in a more personal, relevant and engaging way. Based on this problem, the primary goal became to identify clear customer profiles based on their credit card usage. We tested two possibilities: If our model found clear, well-separated groups (segments with a silhouette score equal to or greater than our determined threshold of 0.5), it would confirm our hypothesis that meaningful segments exist. If not, it would suggest our customers' behaviors are too similar or overlapping to segment effectively.

Data Analysis Process



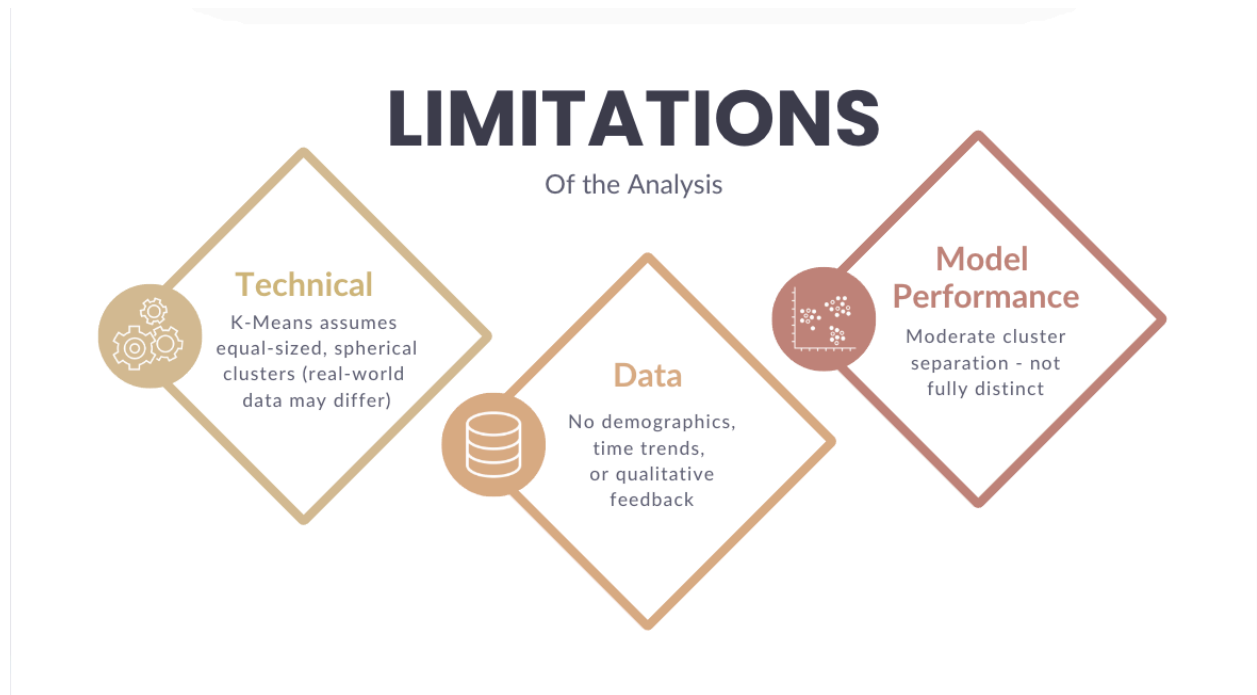
We worked with a dataset of nearly 9,000 credit card customers. It included 18 different measures of behavior - from how often they make purchases, to whether they pay in full, to their use of cash advances. We cleaned the data by filling in missing values, standardizing features, and making sure unusual spending patterns were preserved. EDA (or exploratory data analysis) was then performed to understand the dataset's structure, relationships between variables, and to detect any outliers. Then, we used a technique called K-Means clustering to group customers with similar behaviors. Finally, we used PCA (Principal Component Analysis) to visually check if the segments we identified were clearly distinct.

Findings



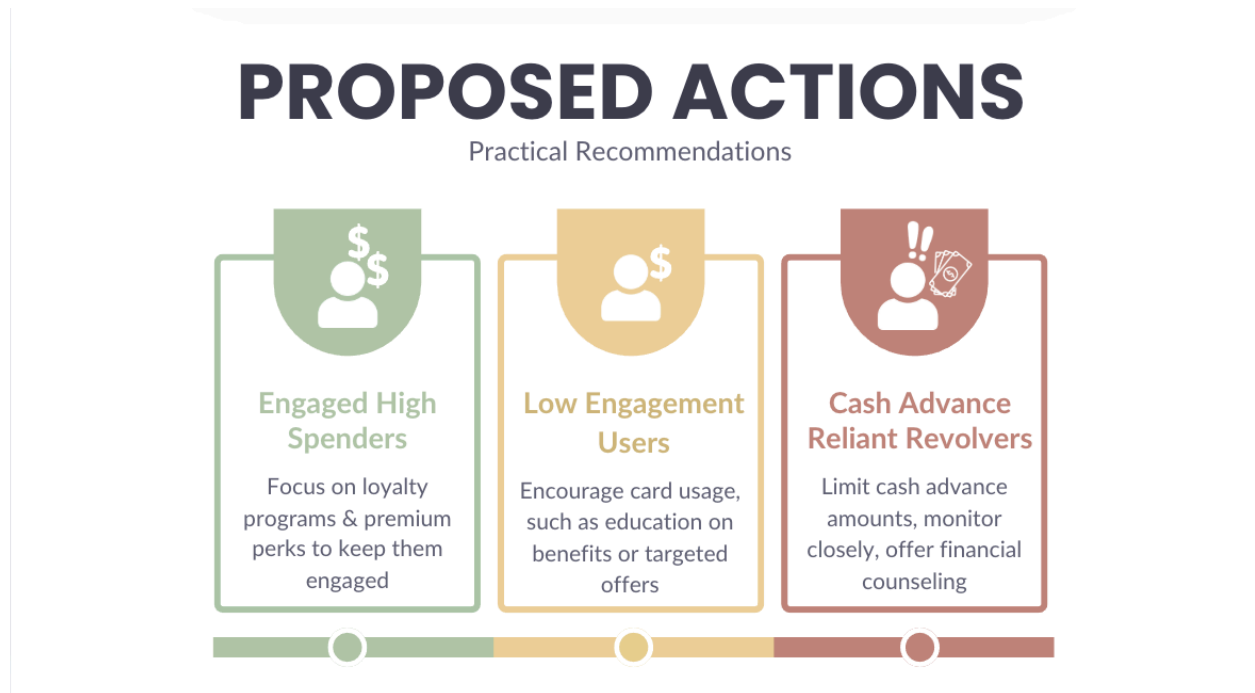
The analysis revealed 3 main customer profiles. First - Engaged High Spenders. These customers make large, frequent purchases and are relatively low risk, as they usually pay in full. Second - Low Engagement Users. They rarely use their cards, possibly because they're new customers or simply prefer other payment methods - leaving their credit cards dormant. And third - Cash Advance Reliant Revolvers. These customers depend heavily on cash advances, carry high balances, and make fewer purchases - putting them in a higher-risk category. The statistical measure of separation between these groups, called the silhouette score, was moderate (at about 0.25), meaning that while one group stood out clearly, the other two clusters overlapped in behavior.

Limitations



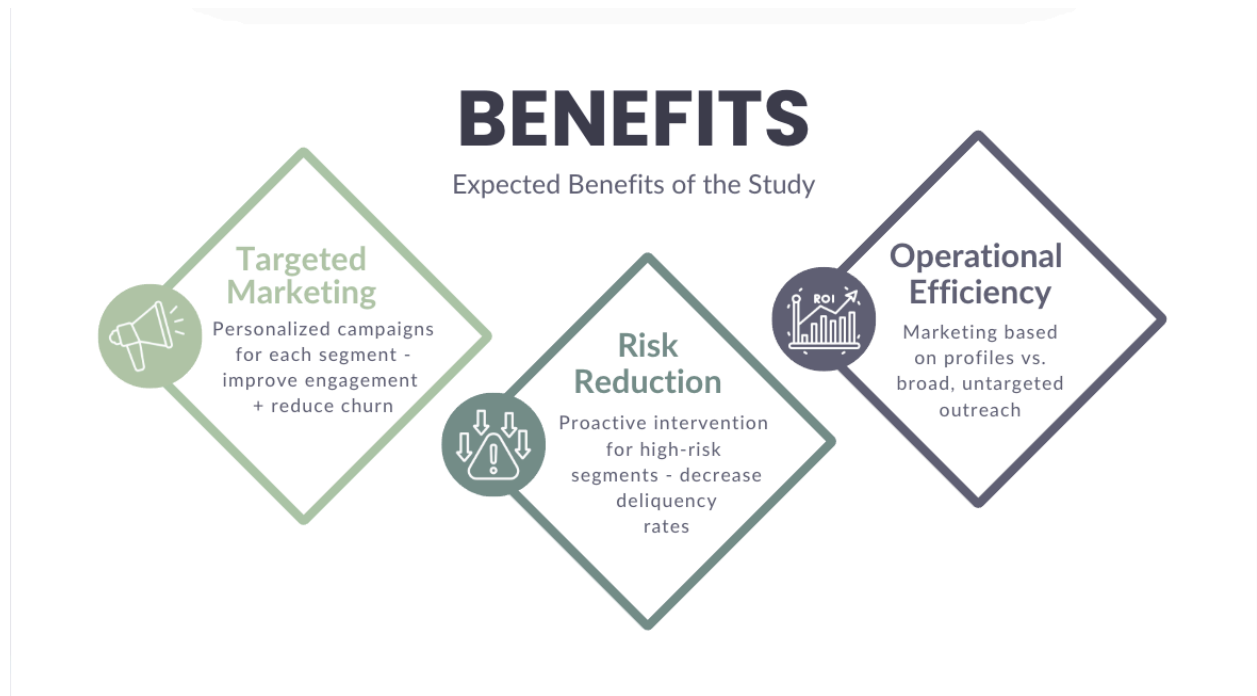
There were some limitations to our approach. Technically speaking, K-Means clustering assumes that all groups are about the same size and shape (GeeksforGeeks, 2025) - unfortunately, this is something that's rarely true with real-world customer data. Our data lacked demographic information and time-based trends, both of which could help sharpen the segmentation. Finally, the moderate silhouette score we mentioned earlier that evaluates our model's performance informs us these clusters aren't perfectly distinct - so any actions based on them should be tested and refined over time.

Proposed Actions



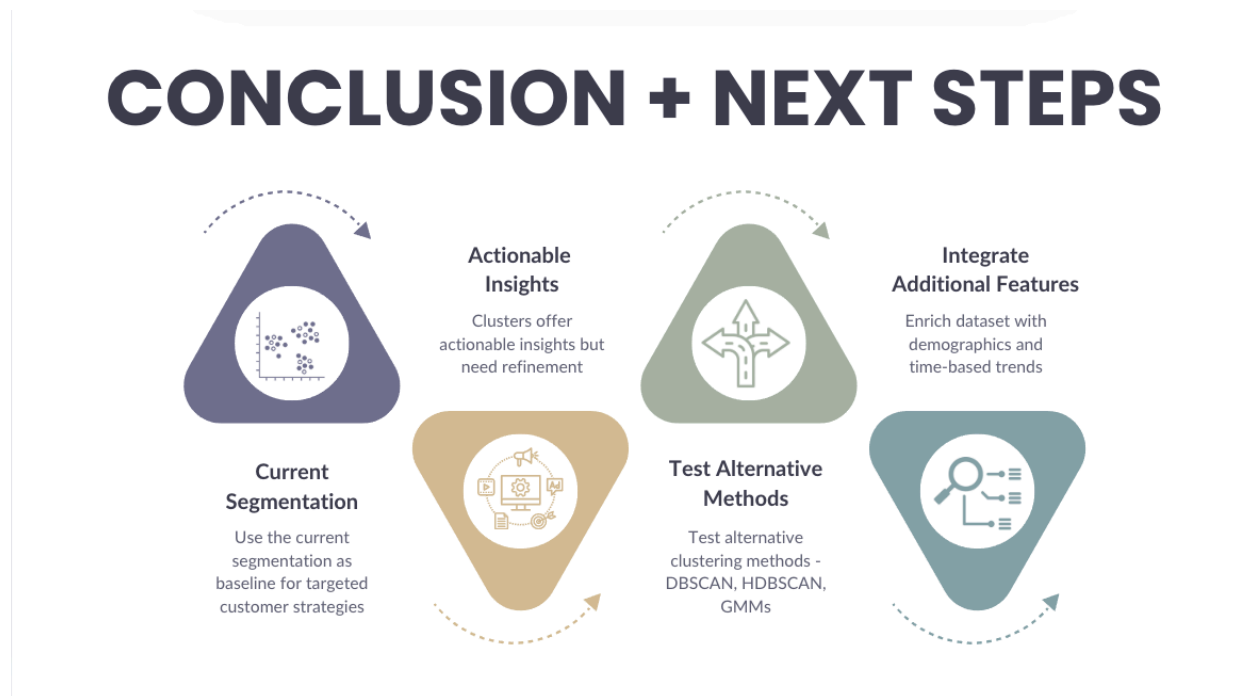
Even with these limitations, the customer profiles we ended up with still give us clear starting points for action. For High Spenders, we can focus on loyalty programs and premium perks to keep them engaged, helping to encourage retention. For Low Engagement Users, we can run marketing campaigns to encourage card usage, such as education on benefits or specifically targeted offers. And for Cash Advance Reliant customers, we can take a risk management approach - such as limiting cash advance amounts, monitoring closely, and offering financial counseling.

Expected Benefits



From industry benchmarks, we know that personalized marketing can increase engagement rates by up to 10-15%, and reduce churn in high-value customers by 5-8% (McKinsey & Company, 2023). Targeted risk reduction strategies can help lower delinquency rates for our higher risk segments. And by focusing our marketing budget where it has the most impact versus casting a wide, untargeted net, we can improve efficiency and ROI overall.

Conclusion



In summary, this segmentation gives us a practical starting point for more focused marketing and risk management. The next step is to refine our model - by testing more advanced clustering methods and adding richer data, additional features like demographic information and seasonal trends. The more accurately we can understand our customers, the more effective and efficient our strategies will become.

References

GeeksforGeeks (July 23, 2025). *Demonstration of K-Means Assumptions*. GeeksforGeeks.

Retrieved August 21, 2025 from

[geeksforgeeks.org/machine-learning/demonstration-of-k-means-assumptions](https://www.geeksforgeeks.org/machine-learning/demonstration-of-k-means-assumptions)

McKinsey & Company. (January 19, 2023). *What is Targeted Marketing? Benefits, Strategies, and Examples*. Salesloft.

Retrieved August 21, 2025 from

salesloft.com/resources/blog/targeted-marketing